

Reflecting on scientific growth and innovation as the Society for Neuro-Oncology turns 30

Macarena I. de la Fuente¹, Priscilla K. Brastianos, Bradley Gampel, Farshad Nassiri, David R. Raleigh², Cristiane M. Ida, Mohamed S. Abdelbaki, Melike Pekmezci, Stephen Bagley, Jacob S. Young, Rimas V. Lukas³, Angela C. Hirbe, Benjamin M. Ellingson, Javier Villanueva-Meyer⁴, Anna Lasorella⁵, Jann N. Sarkaria⁶, David H. Gutmann⁷, Daphne Haas-Kogan⁸, Evanthis Galanis⁹, and Susan M. Chang¹⁰

All author affiliations are listed at the end of the article

Corresponding Author: Macarena I. de la Fuente, M.D., Neuro-Oncology Division, University of Miami, 1120 NW 14th Street, Miami, FL 33136 (mdelafuente@med.miami.edu).

Abstract

The Society for Neuro-Oncology (SNO) marks its 30th anniversary in 2025, providing an opportunity to reflect on scientific advances and future directions in the field. Over 3 decades, SNO has catalyzed scientific innovation, education, mentorship, and global collaboration, advancing the care of patients with primary and metastatic brain tumors. Through its annual meeting and subspecialty conferences in pediatric neuro-oncology and brain metastases, as well as its journals, including *Neuro-Oncology*, *Neuro-Oncology Practice*, *Neuro-Oncology Advances*, and the recently launched *Neuro-Oncology Pediatrics*, SNO has established leading platforms for disseminating knowledge, sharing best practices, and shaping clinical, translational, and basic research worldwide. Scientific milestones during this period include the integration of molecular profiling into central nervous system tumor classification, advances in neuroimaging for diagnosis and treatment monitoring, targeted therapies for selected glioma patients, and the evolution of brain metastases management from whole-brain radiotherapy to multimodal strategies that incorporate targeted and immune-based therapies. Pediatric neuro-oncology has similarly advanced with the use of histomolecular diagnostics, refined risk stratification, and the development of novel targeted agents, alongside an increased emphasis on survivorship. Looking forward, emerging insights into the tumor microenvironment and novel immunotherapeutic approaches offer promising directions for future discovery and translation.

This year marks the 30th anniversary of the Society for Neuro-Oncology (SNO), offering an opportunity to reflect on the multidisciplinary scope of the society and its lasting impact on scientific innovation in neuro-oncology. Over the past 3 decades, SNO has played a crucial role in disseminating scientific knowledge and mentoring the next generation of brain tumor researchers, while creating endless opportunities for networking and international collaboration.

The society membership has increased from 163 in 1995 to over 3000 in 2025, further expanding its role as a multidisciplinary organization, including members from more than 14 disciplines and adding new membership categories, including student, international outreach, and allied health. The annual meeting, along with dedicated stand-alone meetings on pediatric neuro-oncology and central nervous system (CNS) metastases, provides a unique educational, scientific, and networking experience for our global community. With over 2700 participants in 2024, the SNO Annual Meeting is the largest

Neuro-Oncology meeting globally. Additional initiatives to fulfill its mission include the Neuro-Oncology Academy and numerous collaborations with sister scientific societies.

Established in 1999, the Society's flagship publication, *Neuro-Oncology*, with an impact factor of 13.4, is widely recognized as the seminal publication in the field. Alongside its sister journals, and in partnership with the European Association of Neuro-Oncology (EANO) and Oxford University Press, *Neuro-Oncology Practice* (Impact Factor 2.5), focused on applied neuro-oncology, and *Neuro-Oncology Advances* (Impact Factor 4.1), a fully open-access publication, it has served as an ideal platform to share scientific innovation, disseminate best practices, and reach neuro-oncology professionals worldwide. *Neuro-Oncology Pediatrics*, a new SNO/EANO Open Access journal for the pediatric and adolescent young adult neuro-oncology community, was launched in 2025 to showcase the basic, translational, and clinical research and care of this unique patient population, addressing another unmet need.

Received August 29, 2025; accepted October 24, 2025.

© The Author(s) 2025. Published by Oxford University Press on behalf of the Society for Neuro-Oncology. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Over the past 30 years, SNO has emerged as a lightning rod for galvanizing scientific innovation, fostering multidisciplinary global collaboration, mentoring the next generation of brain tumor experts, and disseminating critical knowledge that drives progress in neuro-oncology. In this article, we aim not to reflect on the society's growth but to summarize the most significant scientific discoveries and innovations in the field of Neuro-Oncology as we celebrate the society's 30th anniversary. We invite readers to explore the comprehensive list of references provided in this article for further details on their topic of interest.

Scientific Milestones

Advances in the CNS Tumor Classification

The WHO Central Nervous System Tumor Classification has undergone several revisions in the past 30 years.^{1,2} Multi-omic molecular profiling of CNS tumors dramatically improved our understanding about their genetic and epigenetic basis. Exponential growth of large genomic datasets from whole-exome, whole-genome, and whole-transcriptome profiling along with increasing availability of clinical genetic tests enabled the incorporation of the molecular features into routine CNS tumor diagnostics. More recently, DNA methylation-based classification has emerged as a strategy to refine the classification within well-established tumor types, enable discovery of novel tumor types, and assist in the diagnosis of challenging CNS tumors as an objective, ancillary tool.^{3,4} Molecular features have been progressively incorporated into the diagnostic and grading criteria of multiple CNS tumor types, shifting from a pure histology-based classification up to the 2007 edition to integrated histomolecular-based classification schemes in 2016 and 2021.^{1,2}

Glioblastoma (GBM) was the first CNS tumor type comprehensively evaluated by unprecedented large-scale initiatives that integrated multi-omic and clinical outcome data.^{5–7} These initiatives led to the discovery of IDH mutations, which in conjunction with 1p/19q co-deletion, fundamentally changed the classification framework of diffuse gliomas.^{7–9} Incorporation of IDH mutations and 1p/19q codeletion as diagnostic criteria for adult-type diffuse gliomas consolidated 7 histology-based tumor types in the 2007 classification into 3 histomolecular types in the 2021 classification and excluded the diagnostically controversial “oligoastrocytomas” since 2016. Identification of histone H3 mutations was another pivotal advance in the classification of diffuse gliomas, and inclusion of histone H3 alterations as diagnostic criteria for high-grade pediatric-type diffuse gliomas delineated distinct histomolecular tumor types that were previously grouped as clinicopathologic entities such as diffuse intrinsic pontine gliomas or were misrecognized as CNS primitive neuroectodermal tumors.^{10,11} While it has been long recognized that diffuse gliomas in children have different clinical outcomes than their histologic counterparts in adults, on prior, purely histology-based classifications, they were grouped together with adult tumors. Integration of molecular features also enhanced the classification of low-grade pediatric-type diffuse gliomas, with delineation of distinct tumor types with unifying genomic and epigenetic profiles.^{12–14}

Integration of genetic, cytogenetic, and epigenetic features with histology similarly reshaped the diagnosis and grading of other CNS tumor types. In some cases, multiple, histologically defined tumor entities were unified under a single diagnosis based on their shared molecular features, that is, embryonal tumor with multilayered rosettes and solitary fibrous tumor. In other cases, differences in molecular features enabled recognition of new tumor types among tumors with overlapping histologic features (ie, myxoid glioneuronal tumor and multinodular and vacuolating neuronal tumor).^{15,16} Some purely histologic diagnoses were either modified into more homogenous, integrated diagnoses (ie, astroblastoma, *MN1*-altered and CNS neuroblastoma, *FOXR2*-activated), or were completely replaced by new histomolecularly defined entities (ie, tumors previously classified as CNS primitive neuroectodermal tumors).¹⁷ Ependymal tumors are now classified based on their genetic, cytogenetic, and epigenetic features, which correlate with age, tumor location, and outcome.¹⁸ Further stratification into clinically relevant subtypes/subgroups based on genetic and epigenetic features is enabling personalized treatment decisions, as seen in medulloblastoma^{19,20} and atypical teratoid/rhabdoid tumor.²¹ Molecular features also correlate with patient outcomes and have been incorporated as independent grading criteria for some tumor types, including GBM, IDH-mutant diffuse gliomas, and meningiomas.

As such, we have transitioned to the era of precision diagnostics in neuro-oncology, which expands the opportunity for effective precision therapeutics. This is a rapidly evolving field with emerging CNS tumor types and subtypes/subgroups, and the SNO has been a crucial venue for high-impact studies that are moving this field forward.

Advances in Imaging

Neuroimaging in neuro-oncology has experienced a remarkable transformation over the past 3 decades, revolutionizing brain tumor diagnosis, treatment planning, and monitoring through continuous technological innovations.

Evolution of MRI technology

The 1990s marked a pivotal shift from reliance on CT imaging and basic T1/T2-weighted magnetic resonance imaging (MRI) sequences^{22,23} to sophisticated advanced imaging techniques. Today, 3-Tesla MRI serves as the gold standard for neuro-oncology imaging, with ultra-high-field 7-Tesla systems providing unprecedented resolution and tissue contrast at specialized centers.

Advanced imaging techniques have transformed tumor assessment. Perfusion imaging evaluates blood supply and vessel formation,²⁴ while diffusion imaging²⁵ measures cell density and maps white matter pathways crucial for surgical planning.²⁶ Functional MRI²⁷ identifies critical brain regions to preserve during surgery. Meanwhile, magnetic resonance spectroscopy enables noninvasive metabolic analysis, and emerging hyperpolarized carbon-13 imaging^{28,29} allows real-time visualization of tumor metabolism and treatment response offering unprecedented insights into tumor metabolism and treatment response.

Clinical use of PET imaging

Positron emission tomography (PET) has evolved dramatically from fluorodeoxyglucose-based scanning on PET-only scanners to sophisticated amino acid-based tracers on hybrid PET/MRI scanners. Although no radiolabeled amino acids have been approved by the FDA for diagnostic brain tumor imaging, ^{18}F -FET, ^{11}C -methionine, and ^{18}F -FDOPA have become useful molecular imaging tools, often demonstrating superior tumor delineation and specificity for active disease.³⁰ Recent years have also brought remarkable progress in novel molecular imaging technology. Approaches that were once merely conceptual have now become commercially available, including immunoPET probes for targeted molecular imaging of immune cells and theranostic agents that combine diagnostic and therapeutic capabilities within a single compound.³¹ These advances represent a paradigm shift in personalized neuro-oncology care and are being actively studied.

Clinical trial standardization and the emergence of artificial intelligence

Standardization of MRI protocols marked an important landmark in neuro-oncology imaging. In 2015, Ellingson and colleagues established the first consensus recommendations for brain tumor MRI protocols in gliomas,³² creating a framework for consistent imaging across institutions and clinical trials.³³ This standardization improved reproducibility of imaging findings and facilitated multicenter clinical research. The effort has since expanded to include other primary and metastatic brain tumors,³⁴ as well as other neuro-oncological conditions³⁵ and advanced imaging applications,³⁶ improving clinical research and patient care through more consistent and comparable imaging assessments.

Alongside these protocol standardization efforts, response assessment methodologies also evolved. The Response Assessment in Neuro-Oncology (RANO) criteria,³⁷ established in 2010, marked a watershed moment in standardizing tumor measurement protocols across clinical trials and practice. The recent RANO 2.0 update further refines these criteria,^{38,39} enhancing our ability to identify promising therapies and assess treatment response more accurately.

Building on these efforts, once labor-intensive manual tumor measurements are now being replaced by artificial intelligence-powered automated volumetric segmentation,^{40,41} providing rapid and reproducible tumor volume quantification for objective response assessment. Continued advances in artificial intelligence and increased data sharing will unlock artificial intelligence's potential to enhance imaging-based diagnosis, prognostication, and response assessment for neuro-oncology.⁴²

Advances in Surgical Neuro-Oncology

Surgical neuro-oncology has undergone transformative progress over the past 3 decades, largely driven by innovations in imaging, intraoperative technologies, and a more sophisticated understanding of brain tumor biology.

Innovations in imaging, surgical techniques, and functional monitoring have enabled greater precision, improved preservation of function, and safer resections. High-resolution structural MRI, functional MRI, diffusion tensor imaging, and metabolic imaging with PET now allow for detailed mapping

of the tumor in relation to white matter tracts and eloquent brain. These advanced imaging modalities can integrate with frameless stereotactic navigation systems to provide real-time 3-dimensional intraoperative spatial localization, while other adjuncts such as intraoperative MRI and ultrasound can provide enhanced visualization of residual tumor during surgery to support more complete resections.

Operative techniques have also evolved in parallel, with an increasing focus on minimizing invasiveness without compromising oncologic outcomes. Advances in our understanding of cortical and subcortical electrophysiology have enabled continuous monitoring of functional pathways—both in awake and asleep procedures—supporting safe resection even in eloquent regions. Fluorescence-guided surgery, for example with 5-aminolevulinic acid (5-ALA), is increasingly being used for resection of high-grade gliomas to enhance intraoperative tumor visualization and improve extent of resection. Even tumors that were once considered surgically inaccessible can now be treated using minimally invasive image-guided tumor ablative strategies, for example, using laser interstitial thermal therapy. Some tumors, particularly those that are often found at the base of the skull, are now routinely resected through approaches that do not transgress the cranium, offering access without a craniotomy and reducing morbidity.

Importantly, modern surgical neuro-oncology has been reshaped by molecular and genetic insights, making a shift toward biology-informed and personalized surgeries. Rapid intraoperative molecular diagnostics are emerging, enabling real-time decisions that may influence surgical strategy or guide suitability for targeted therapies. Locoregional therapies are also an expanding frontier. Delivery of biologics, oncolytic viruses, or chemotherapeutics, either directly into the tumor or by convection-enhanced delivery, bypasses the blood-brain barrier to target the tumor directly and potentially modulate the microenvironment.

Collectively, these innovations have enhanced surgical precision, functional preservation, and in some tumors, survival. Together these trends define the neurosurgeon's evolving role at the intersection of surgery, biology, and translational neuro-oncology.

Advances in Non-Invasive Technology

Tumor treating fields

Tumor treating fields (TTF) have emerged as a novel noninvasive therapeutic modality in the treatment of GBM, grounded in advances in bioelectromagnetics. TTF delivers low intensity, alternating electric fields that are thought to disrupt mitosis in dividing tumor cells while largely sparing normal brain tissue. When added to maintenance temozolomide, TTF significantly improved overall survival in newly diagnosed GBM in a randomized controlled trial (20.9 m, TMZ + TTF versus 16.0 m, TMZ). Ongoing studies are evaluating its use in other malignancies and in combination with other treatments such as checkpoint inhibitors in GBM.

Advances in Radiation Therapy

Over the past 30 years, the field of radiation oncology has become inherently dependent on evolving technology and

computing power. In 1995, radiation therapy planning involved obtaining 2-D planar X-ray images and then manually transposing tumor volumes and critical organs at risk from CT (or, less commonly, MRI) scans onto “simulation” films, often with a grease pencil. Radiation planning was performed at multiple levels in 2 dimensions, and strategies to improve homogeneity within radiation fields were limited to using brass wedges to attenuate beams. Radiation fields on linear accelerators (LINACs) were shaped using lead alloy blocks that were fabricated manually, a laborious process that limited throughput and made on-treatment re-planning (when necessary) challenging if not impossible. Stereotactic treatments using the Gamma Knife system were commonplace, but LINAC-based radiosurgery was limited to a few academic centers, and both techniques allowed treatment of only one tumor target at any one time. The concept of intensity-modulated radiation therapy (IMRT) was just being tested using the PEACOCK system, and proton treatment centers were limited to a handful of institutions, such as Loma Linda University and Massachusetts General Hospital.

In contrast to the technological and computing standards of 1995, radiation therapy in 2025 is vastly different. Modern treatment planning systems use powerful software to align CT simulation scans with diagnostic MRI and PET imaging studies to allow facile delineation of tumor and normal tissues in 3-dimensions. Instead of static lead blocks, radiation fields are shaped by mechanized metal leaflets (eg, multileaf collimators) that are integrated into LINACs. Importantly, multileaf collimators move dynamically during radiation delivery to enable exquisite modulation of delivered dose across radiation fields that are individualized to each patient, sometimes on a daily basis. This capability is fundamental to IMRT, a radiation planning approach that was in its infancy in 1995 but is now standard-of-care. IMRT has critically enhanced our ability to limit radiation dose to critical tissues, such as the optic nerve, brain stem, and hippocampus, while still delivering higher doses of radiation to tumor targets than were possible in 1995. The advent and improvements in IMRT now allow simultaneous radiosurgery for many brain metastases, which has revolutionized care for patients and is increasingly making whole brain radiation therapy uncommon. The proliferation of proton centers across the country has similarly allowed for the development of highly conformal radiation plans that can further limit the dose of radiation to normal tissues; maturation of clinical trials comparing different RT modalities is expected to strengthen evidence-based guidelines.

Advances in Systemic Therapies in Primary Brain Tumors

The dramatic improvement in our understanding of molecular drivers of pathogenesis in CNS tumors has enabled not only a more accurate tumor classification but also impacted therapeutic opportunities and outcomes, in selected groups of patients, particularly during the last decade.^{2,5-7} However, despite these advances, median overall survival for GBM, the most common malignant brain tumor, remains poor at 14–16 months with no significant improvement over the last decades.

Temozolomide, an oral alkylating agent, has been the backbone of systemic therapy for GBM and other gliomas since 2005, based on the results of an international phase III trial that demonstrated its statistically significant survival benefit in combination with radiotherapy.⁴³ A simultaneous publication showed that this benefit was primarily driven by MGMT promoter methylated GBM with marginal benefit for unmethylated GBM. More recently, the CATNON trial confirmed the survival benefit of adjuvant temozolomide for grade 3 astrocytoma.⁴⁴ The addition of procarbazine, lomustine, and vincristine (PCV) to radiation therapy in WHO grade 3 oligodendroglioma increases both overall survival and progression-free survival. This regimen has been used in grade 2 and grade 3 oligodendrogliomas and grade 2 astrocytoma interchangeably with temozolomide based on patient performance and physician preference.^{45,46} Bevacizumab, a recombinant humanized monoclonal antibody against human vascular endothelial growth factor was approved in 2009 as single agent for patients with recurrent GBM. The approval was based on clinically meaningful and durable objective response data from 2 studies.^{47,48} Although bevacizumab has failed to demonstrate a survival benefit in the upfront or recurrent setting for GBM, it has a significant role in the treatment of symptomatic patients, patients who are ineligible for cytotoxic chemotherapy, and radiation necrosis.

The identification of *IDH* mutations in gliomas and the efforts to target those alterations represent one of the major hallmarks in the field of neuro-oncology over recent decades. Following the results of the practice-changing INDIGO trial, vorasidenib, a brain-penetrant small molecule inhibitor of mutant *IDH1/2* proteins, received regulatory approval for WHO grade 2 *IDH*-mutant glioma following surgery, including biopsy. This has been followed by regulatory approvals in other countries.⁴⁹

Targeted therapies may provide effective and durable therapeutic options for highly selected patients with primary brain tumors harboring less common but actionable molecular alterations, such as *BRAF*, *FGFR*, or *NTRK* alterations. The most compelling data emerge from targeting the *RAS*-*RAF*-*MEK*-*ERK* pathway. Dabrafenib/trametinib (*BRAF*/*MEK* inhibitors) combination received FDA approval as a tissue-agnostic treatment for advanced solid tumors harboring *BRAF*^{V600E} mutation, including adult and pediatric low- (LGG) and high-grade gliomas (HGG), and for first-line treatment of patients with pediatric *BRAF*^{V600E} LGG who require systemic therapy.^{50,51} Furthermore, *BRAF*-*MEK* inhibition showed impressive results in newly diagnosed papillary craniopharyngiomas, with 94% of patients achieving a durable partial response or better.⁵² Tovorafenib, a first-in-class *BRAF* type II inhibitor, was recently approved for patients 6 months of age and older with relapsed or refractory pediatric LGG harboring a *BRAF* fusion or rearrangement, or *BRAF*^{V600} mutation.⁵³

Dordaviprone (ONC201) an oral, blood-brain barrier penetrant, antagonist of dopamine receptor D2/3 and allosteric agonist of the mitochondrial protease caseinolytic mitochondrial matrix peptidase proteolytic subunit, recently received accelerated approval by the Federal Drug Administration for recurrent H3 K27M-mutant diffuse midline glioma. This was based on the data on 50 patients across 4 clinical trials and

a compassionate use program. ORR was 20% and median duration of response was 11.2 months.⁵⁴

Although the progress over the last decades has been remarkable, considerable work remains to impact patient survival. Key priorities include preclinical efforts to identify more homogeneous subgroups of patients with actionable alterations who are most likely to benefit from these therapies, as well as the development of innovative clinical trial designs and multicenter collaborations to ensure clinical trial feasibility in rare patient populations.

Advances in the Treatment of Neurogenetic Disorders

Over the past 30 years, advances in treating several rare neurogenetic disorders have emerged, built on decades of preclinical and clinical research highlighting the impact of such work on patients' lives. In diseases unified by germline mutations in single driver genes, targeted small molecule therapies have proven particularly effective. We highlight 2 examples.

The common benign peripheral nerve sheath tumors, termed plexiform neurofibromas, cause substantial morbidity for patients with neurofibromatosis type 1 (NF1). Loss of *NF1* gene expression in these tumors results in increased RAS-ERK pathway activation, such that MEK1/MEK2 blockade is associated with objective radiographic responses and improvement in pain. Two MEK1/MEK2 inhibitors now have FDA approval, selumetinib for children⁵⁵ and mirdametinib for adults and children.⁵⁶

Von Hippel-Lindau disease (VHL), driven by *VHL* mutations increases HIF-2 α activity leading to the development of CNS hemangioblastomas. Targeting HIF-2 α with belzutifan has demonstrated a decrease in solid and cystic components of CNS hemangioblastomas and has decreased the need for invasive interventions.⁵⁷

SNO has catalyzed this field by fostering collaboration among clinicians, basic researchers, and translational investigators. The creation of a dedicated Neurogenetic Tumor Syndrome track at the annual meeting and a full Education Day (2023) are 2 key examples. For rare diseases with unifying themes, this support mechanism has proven invaluable and promises even greater future impact.

Advances in Meningioma

Our understanding of the molecular underpinnings of meningioma has dramatically increased in the last 15 years, starting with the first comprehensive genomic characterization of meningiomas in 2 seminal papers in 2013.^{58,59} These papers were among the first to reveal potentially clinically actionable alterations in meningiomas, including *AKT* and *SMO* mutations. These studies also set the stage for the hypothesis that genetic alterations correlate with clinical phenotypes in meningiomas. For example, *AKT*, *PIK3CA*, and *SMO* mutations tend to occur in tumors of the skull base, *BAP1* alterations in rhabdoid meningiomas, and *CDKN2A/B* homozygous loss and *TERT* promoter mutations are now incorporated into the WHO guidelines as defining grade 3

meningioma. More recently, multi-omic approaches, including gene expression and DNA methylation analyses, have refined molecular subclasses of meningioma.^{60–66} With these molecular advances, there are ongoing biomarker-driven trials for patients with recurrent/progressive meningioma after standard surgery and/or radiation, which will elucidate the role of targeted therapy in meningiomas.⁵⁷

In addition to evaluating targeted therapies in meningiomas, there is interest in the field to investigate immunotherapeutic approaches, as meningiomas harbor an immunosuppressive environment and express PD-L1.^{68,69} Unfortunately, those trials have shown mixed results. Thus, more work is needed to identify which subset of patients will benefit from immune checkpoint blockade. Additional studies are evaluating combinations of immunotherapy with other agents, including radiation.

Finally, since meningiomas express somatostatin receptor type 2 (SSTR2), peptide receptor radionuclide therapy targeting SSTR2 (177-Lu-DOTATE) is now being investigated as a potential novel therapeutic approach for meningiomas.⁷⁰

In summary, although surgery and radiation continue to remain the cornerstone of therapy for patients with meningioma, systemic therapy is emerging as playing a more meaningful role for patients with recurrent/progressive meningiomas.

Advances in CNS Metastases

The management of brain metastases has changed dramatically in the last 30 years. Whereas historically, most patients with brain metastases received whole-brain radiation as the mainstay of treatment, improvements in radiation and surgical techniques as well as systemic therapies have altered the therapeutic landscape of this disease.⁷¹ Indeed, in this era of precision oncology, systemic therapies, most notably immunotherapy and targeted therapy, are playing an increasingly important role in the management of patients with brain metastases.

In non-small cell lung cancer, therapies targeting alterations such as EGFR, ALK, NTRK, and KRAS have shown significant intracranial activity in patients with metastatic cancer.^{72–75} For patients with advanced HER2-amplified breast cancer, where brain metastases occur in about 30%–40% of patients, drugs targeting HER2 are becoming standard for patients with brain metastases, particularly, newer HER2-agents, such as tucatinib, when combined with trastuzumab and capecitabine.⁷⁶ Antibody-drug conjugates (ADCs), which consist of a monoclonal antibody linked to a cytotoxic payload, selectively deliver drug to antigen-expressing cells. ADCs, such as trastuzumab and deruxtecan, are showing promising efficacy in patients with brain metastases and are being investigated in trials. Melanoma has seen similar exciting progress, with BRAF and MEK inhibitors being active in patients with BRAF-mutant brain metastases.⁷⁷ Immune checkpoint inhibition has also shown responses in >50% of patients with brain metastases,^{78,79} which is a dramatic improvement when compared to outcomes more than 15 years ago. Based on these data, in situations where there are active systemic agents, upfront systemic therapy can be considered in patients with asymptomatic brain metastases when local control with surgery or radiation is not urgently needed.

Although these data are promising, significant work is still needed to move the needle in the area of brain metastases. Only a subset of patients responds to therapy, and the majority of patients develop resistance to therapy. Recent studies have shown that brain metastases evolve genomically from their primary tumors and have clinically actionable alterations that may not be detected in the clinically sampled primary tumor.^{80,81} The blood-brain barrier and the blood-tumor barrier limit drug delivery. For this reason, investigations into the identification of therapies that target drivers of metastasis and that adequately penetrate the brain will be critical for the field moving forward.

Strategies to reduce cognitive deficit and prevent negative impact of radiation therapy in brain metastases included increased use of focal delivery of radiotherapy, tailored approaches such as hippocampal-avoidant whole-brain radiotherapy (WBRT), and the use of memantine as neuro-protectant during WBRT.

Advances in Pediatric Neuro-Oncology

Over the past three decades, the field of pediatric neuro-oncology has transformed dramatically. The standard for classifying tumors now involves an integrated approach combining histological and molecular diagnoses. Treatment and prognosis are determined not just by the type of tumor but by the specific mechanisms driving tumor growth.⁸² DNA methylation profiling has improved our ability to differentiate CNS tumors,⁸³ which was reflected in the latest 2021 WHO CNS classification that introduced 22 new disease entities while removing others such as primitive neuroectodermal tumor (PNET).² The expertise of pediatric neurosurgeons in obtaining a biopsy from delicate areas in the brain has been an integral component of advancing the field.⁸⁴ Translational research from brainstem tissue has provided hope for previously incurable conditions like Diffuse Intrinsic Pontine Glioma.⁸⁵ Liquid biopsies that detect tumor DNA in cerebrospinal fluid (CSF) have emerged, offering potentially less invasive methods for diagnosis and surveillance.⁸⁶ Medulloblastoma now has four distinct molecular groups, enabling therapy de-escalation for WNT-activated group, and future intensification for high-risk diseases, such as *SHH*-activated, *TP53* mutant group, and Group 3 with *MYC* amplification and/or isochromosome17q. Imaging of pediatric CNS tumors has evolved with the realization that pediatric patients require different imaging techniques than adults. The establishment of the Response Assessment in Pediatric NeuroOncology (RAPNO)^{87–89} has begun standardizing protocols for all pediatric CNS tumors, facilitating better comparisons of clinical trials and treatments globally. A significant change includes the reliance on T2 (fluid-attenuated inversion recovery) FLAIR images in assessing the response for pediatric low-grade gliomas (pLGGs) compared to depending on post-contrast imaging.⁹⁰ The improved ability to characterize tumors has led to more individualized therapies. Targeted agents such as Dabrafenib/Trametinib, Tovorafenib, and Vorasidenib have become the standard of care in the newly diagnosed and recurrent settings, with numerous others under evaluation.^{91–93} Promising multi-modality treatments, including various immunotherapies and methods to surpass the

blood-brain barrier, have been developed. Efforts are also focused on minimizing long-term toxicities and maximizing quality of life. Proton beam therapy has become a preferred radiation therapy (RT) modality for children with brain tumors, allowing targeted delivery to reduce RT exposure to healthy brain tissue.⁹⁴ As our ability to treat CNS tumors has improved, so has our understanding of survivorship. We continue to learn new ways to manage posttreatment cognitive deficits, hormonal imbalances, and organ damage.⁹⁵ Additionally, pediatric-type tumors are now recognized as occurring throughout early adulthood, prompting the SNO to establish an Adolescent and Young Adult (AYA) working group. Several ongoing pediatric clinical trials now include patients up to 39 years of age.

Over the past 30 years, pediatric neuro-oncology has fundamentally shifted from a one-size-fits-all model to recognizing the unique biology of pediatric tumors distinct from their adult counterparts. Precision medicine now tailors each patient's treatment to their individual tumor, with an expanded emphasis on not only curing diseases but also prioritizing long-term quality of life.

Current Investigations and Future Directions

Immunotherapy and Cellular Therapy

Immunotherapy, predominantly in the form of immune checkpoint inhibition, has revolutionized the care of numerous malignancies over the past 15 years. However, immune checkpoint inhibitors have thus far failed to demonstrate efficacy in patients with primary brain cancers, including GBM.^{96–98} While discovery of mechanisms of resistance and attempts to sensitize glial tumors to immune checkpoint inhibition remain active areas of investigation, a wide array of alternative approaches are currently under investigation.

One area of ongoing interest is virotherapy and immunovirotherapy, a form of immunotherapy in which oncolytic-virus-induced oncolysis alters the tumor microenvironment (TME), leading to activation of proinflammatory pathways and recruitment of immune cells through exposure of viral and tumor antigens. Many viruses have been evaluated in preclinical models and early-phase clinical trials. Unfortunately, response rates have generally been low across phase 1 clinical studies reported thus far, direct intratumoral injections of modified versions of adenovirus,⁹⁹ poliovirus,¹⁰⁰ herpes simplex virus,¹⁰¹ and measles.¹⁰² Vaccines have also generated intriguing results, including peptide, dendritic cell, DNA, and RNA vaccines. Novel approaches continue to be explored, including CMV-specific dendritic cell vaccines, combinations with immune checkpoint inhibition, and direct injection near the deep cervical lymph nodes. DNA plasmid vaccines and mRNA vaccines remain in early-phase trials for GBM but have already been shown to generate glioma-specific immune responses and potential early efficacy signals.^{103,104}

In addition to oncolytic viral therapies and vaccines, cellular therapies have also gained traction over the past few years. Intracerebroventricular administration of chimeric antigen receptor (CAR) T cell therapies is being evaluated

across both pediatric and adult high-grade gliomas. Some early efficacy signals have been observed in children with diffuse midline glioma (GD2-targeted and B7H3-targeted) and adults with recurrent GBM (IL13 α 2-targeted cells,¹⁰⁵ EGFRvIII-targeted cells that secrete a T cell-engaging antibody molecule against wild-type EGFR,¹⁰⁶ and bivalent CAR T cells targeting EGFR epitope 806 and IL13 α 2 [ref.¹⁰⁷]) Longer follow-up time and eventual randomized clinical trials are needed to confirm these encouraging early results.

Drug Delivery with Focused Ultrasound

Low-frequency focused ultrasound (FUS) is an emerging strategy to transiently disrupt the blood-brain barrier, enhancing drug delivery in gliomas. Two modalities, MRI-guided FUS and implantable US devices, have been demonstrated to be feasible and safe in early-phase studies, and their efficacy is currently being tested in phase 3 studies. Beyond traditional chemotherapy, preclinical models of glioma have explored focused ultrasound as a means for improving CAR T-cell delivery,^{108,109} liponanoparticles,¹¹⁰ and even larger therapies, such as antibodies and ADCs.¹¹⁰ Exciting, cutting-edge bioengineering approaches are being used to facilitate the tailored delivery of a therapeutic payload directly in the sonicated area, including sonication-mediated CAR T-cell activation and ultrasound-responsive particles carrying a therapeutic payload. Ultrasound-mediated delivery of chemotherapy can be immunomodulatory, with emerging preclinical data indicating that sonication itself has immune-altering effects that increase immune cell trafficking/recruitment and facilitate an immunogenic cell death,^{111,112} which has been shown to provide a synergistic effect when combined with checkpoint inhibition¹¹³ and may also improve responses to other immunotherapies. Finally, FUS technologies have emerging applications beyond drug delivery and are being utilized to augment liquid biopsy sensitivity and induce cancer cell death as a sonodynamic therapy when combined with aminolaevulinic acid or other chemotherapies.

The Role of the Tumor Microenvironment in Gliomas and Cancer Neuroscience Therapeutics

Over the past few decades, neuro-oncology has been reimagined as a neurological disorder in which brain cancer initiation, growth, and progression are dependent upon non-cancerous cells in the TME. In this regard, gliomas are composed of numerous distinct non-neoplastic cell types, including tumor-associated monocytes,¹¹⁴ T lymphocytes,^{115,116} and neurons.^{117,118} As the relationships between each of these stromal cell types and glioma cells have been defined, new opportunities for cancer therapeutics have emerged.

Numerous studies have revealed that tumor cells markedly influence microglia/macrophage function relevant to adult and pediatric glioma growth.^{119–121} Leveraging these findings, new therapies that deplete myeloid cell recruitment (CSF1R inhibition), block immune modulator function, or target immune checkpoint proteins (TIM-3) have been evaluated in preclinical models of LGG and HGG.^{122–126} In

addition, while T lymphocytes represent a small cellular component of most LGG and HGG,^{127,128} they are important cellular targets for immune modulatory therapies.^{129–131}

Neurons also form productive relationships with brain tumor cells, dictating their growth through indirect interactions with other cell types (eg, oligodendrocyte precursor cells, T lymphocytes)^{132–135} or via direct neurotransmitter receptor or gap junction coupling. These cancer neuroscience insights have already resulted in translational efforts aimed at silencing neuron–glioma interactions by repurposing anti-epileptic drugs to suppress brain cancer growth.¹³⁶

Summary and Conclusions

Marking its 30th anniversary, the SNO has been a transformative force in advancing neuro-oncology research, education, and global collaboration. Over 3 decades, SNO has catalyzed progress through its annual meeting, subspecialty conferences, and leading publications, fostering scientific innovation and mentoring new generations of brain tumor experts while providing educational platforms like the SNO Academy. Major scientific milestones include the integration of molecular profiling into CNS tumor classification, which has not only improved diagnostic accuracy but also enabled transformative precision medicine approaches in selected groups of patients. Advances in neuroimaging have further enhanced tumor detection, characterization, and treatment monitoring. The management of brain metastases has shifted from whole-brain radiation to multimodal approaches incorporating targeted therapies and immunotherapies, significantly improving intracranial control. Pediatric neuro-oncology has similarly evolved through integrated histomolecular diagnoses, refined risk stratification, and novel targeted agents, with survivorship care focused on quality of life. Immunotherapy approaches, including immune checkpoint inhibitors, oncolytic viruses, vaccines, and CART cells, are promising but require further optimization, especially for primary brain tumors. Growing insights into the TME reveal complex interactions among glioma cells, immune cells, and neurons, opening new therapeutic avenues targeting noncancerous stromal elements.

Looking ahead, continued efforts are needed to translate these scientific breakthroughs into improved patient outcomes, as median survival has not significantly changed for GBM, diffuse midline glioma, and many other CNS tumors. Future directions involve refining molecular diagnostics to identify more homogeneous patient subgroups with actionable alterations, facilitating personalized therapies. Innovative clinical trial designs and expanded multicenter collaborations will be critical to overcoming challenges in studying rare tumor populations. Additionally, emerging priorities in neuro-oncology include quality of life, neurocognitive preservation, survivorship, and caregiver support. Embedding these endpoints in clinical trials will ensure therapeutic advances yield meaningful, patient-centered outcomes. Furthermore, leveraging artificial intelligence and advanced imaging modalities promises to improve diagnostic accuracy and therapeutic response assessment. As neuro-oncology rapidly evolves, SNO's role as a global collaborative platform will remain essential in advancing

innovation, fostering collaboration, and enhancing education and mentorship to impact patients' outcomes worldwide.

Keywords

brain metastases | glioma | pediatric neuro-oncology | Society for Neuro-Oncology (SNO)

Funding

There was no manuscript-targeted external funding. Outside of the submitted work, M.I.D. was supported in part by The Sylvester Comprehensive Cancer Center Support Grant 5P30CA240139-04, The Dowskin Family Foundation, and by National Institutes of Health (NIH) grants 2UM1-CA186644-06, 1R37CA262510-01A1, and 1R21CA282543.

Acknowledgments

The authors thank Carol Kruchko, Vinay Puduvalli, Monica Venere, Chas Haynes, and Shelley Pressley for their contribution to the SNO History Committee.

Conflict of Interest Statement

The text is the sole product of the authors. No third party had input or gave support to its writing. M.I.D.: Advisory Board/Consultant: Anheart, Fore, Rigel, Nuvation, Servier. Honoraria for role as speaker: Medscape. Clinical trial support to University of Miami from MSKCC, Servier, Fore, Peloton, and GCAR. Research support from Caris. P.K.B.: grants from: NIH, National Brain Tumor Society, Melanoma Research Alliance, Kinnate (institutional grant to MGH), Breast Cancer Research Foundation, Merck (institutional grant to MGH), American Association for Cancer Research, Eli Lilly (institutional grant to MGH), MGH ECOR Scholars Program (Terry and Jean de Gunzburg Research Scholar), Mirati (institutional grant to MGH), Demetra Fund (Hellenic Women's Association), Krantz Award. Consulting fees from: Atavistik, CraniUS, Exelixis, Genentech, InCephalo, Kazia, Medscape, Selectin, and MPM Capital Advisors. Payment for lectures from Medscape. Stock (Selectin). Clinical trial support (to institution) from Kazia, AstraZeneca, Merck, Pfizer, Eli Lilly, GlaxoSmithKline, Mirati, Genentech-Roche, Tesaro, and Bristol Myers Squibb; Medical Writing (Eli Lilly). B.G.: nothing to disclose. F.N.: nothing to disclose. D.R.R.: nothing to disclose. C.M.I.: nothing to disclose. M.S.A.: grants from CureSearch Catapult Grant, Rally foundation, and National Cancer Institute (NCI). Consulting fees and honoraria from IPSEN Biopharmaceuticals and Day One Biopharmaceuticals. Participation on a Data Safety Monitoring Board or Advisory Board, Day One Biopharmaceuticals, CTI-Clinical Trial Services, Inc. M.P.: royalties, Elsevier, Inc. Stock, Ocugen. S.B.: grants/contracts: Kite, GSK, Lilly, Novocure. Incyte. Consulting fees: Servier,

Chimerix, Telix, and Modifi Bio. Payment for lectures/presentation, Physicians' Education Resource. J.S.Y.: grants from K12CA260225—scholar, UCSF Brain Tumor SPORE Career Enhancement Program. Planned patent: Dyrk1a inhibitor patents planned but not yet submitted. R.V.L.: grant from NIH. Consulting fees from Merck, Novocure, Servier, and Telix. Honoraria for lecture/presentations from Merck, Novocure, and Servier. Drug only for clinical trial, BMS. A.C.H.: nothing to disclose. B.M.E.: grants/contracts: Siemens and Neosoma to UCLA. Consulting fees (self), Autem, Chimerix Inc, Global Coalition for Adaptive Research (GCAR), Imaging Endpoints, Medicenna, Voiant, Medscape, LLC, MeritCRO, Monteris, Neosoma, Nerviano Medical Sciences, Nuvation Bio, Orbus Therapeutics, Sagimet Biosciences, Sapience Therapeutics, Servier Pharmaceuticals, Siemens, SonALAsense, Sumitomo Dainippon Pharma Oncology, Telix, and Third Rock Ventures. Data Safety Board, Carthera. J.V.M.: nothing to disclose. A.L.: nothing to disclose. J.N.S.: nothing to disclose. D.H.G.: grants from Gilbert Family Foundation (Vision Restoration Initiative to D.H.G.), and National Institute of Neurological Disorders and Stroke (1R35NS142919-01 to D.H.G.). Patent pending for the use of lamotrigine for NF1 Brain Tumors. D.H.K.: Consulting fees, Gerson Lehman Group, Guidepoint Global. Honoraria for lecture/presentations, IBA Proton Therapy Inc. Cincinnati Children's Hospital DSMB. Equipment/materials/drugs, Artios Pharma. E.G.: grants/contracts Servier and Denovo to Institution. DSMC, Boston Scientific, Servier Pharmaceuticals, Boehringer Ingelheim, and Modifi Biosciences. S.M.C.: nothing to disclose.

Affiliations

Sylvester Comprehensive Cancer Center, University of Miami, Miami (M.I.D., B.G.); Department of Neurology, University of Miami, Miami (M.I.D.); Department of Medicine, Harvard Medical School, Boston (P.K.B.); Department of Pediatrics, University of Miami, Miami (B.G.); Department of Surgery, University of Toronto, Toronto (F.N.); Department of Radiation Oncology, University of California, San Francisco (D.R.R.); Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester (C.M.I.); The Division of Hematology, Oncology, and Bone Marrow Transplant, St Louis Children's Hospital, Washington University School of Medicine, St Louis (M.S.A.); Department of Pathology, University of California, San Francisco (M.P.); Department of Medicine, University of Pennsylvania, Philadelphia (S.B.); Department of Neurosurgery, University of California, San Francisco (J.S.Y., J.V.M.); Department of Neurology, Northwestern University, Chicago (R.V.L.); Division of Oncology, Washington University, St. Louis (A.C.H.); UCLA Brain Tumor Imaging Laboratory, Department of Radiological Sciences, David Geffen School of Medicine, University of California, Los Angeles, Los Angeles (B.M.E.); Department of Radiology and Biomedical Imaging, University of California, San Francisco (J.V.M.); Department of Biochemistry, University of Miami, Miami (A.L.); Department of Radiation Oncology, Mayo Clinic, Rochester (J.N.S.); Department of Neurology, Washington University School of Medicine, St. Louis (D.H.G.); Department of Radiation Oncology, Dana Farber Cancer Institute, Boston; Department of Oncology, Mayo Clinic, Rochester (E.G.); Division of Neuro-Oncology, Department of Neurosurgery, University of California, San Francisco (S.M.C.)

References

- Louis DN, Perry A, Reifenberger G, et al. The 2016 world health organization classification of tumors of the Central nervous system: a summary. *Acta Neuropathol.* 2016;131:803–820.
- Louis DN, Perry A, Wesseling P, et al. The 2021 WHO classification of tumors of the central nervous system: a summary. *Neuro Oncol.* 2021;23:1231–1251.
- Capper D, Jones DTW, Sill M, et al. DNA methylation-based classification of central nervous system tumours. *Nature.* 2018;555:469–474.
- Aldape K, Capper D, von Deimling A, et al. cIMPACT-NOW update 9: recommendations on utilization of genome-wide DNA methylation profiling for central nervous system tumor diagnostics. *Neurooncol Adv.* 2025;7:vdae228.
- Cancer Genome Atlas Research Network: McLendon R, Friedman A, Bigner D, et al. Comprehensive genomic characterization defines human glioblastoma genes and core pathways. *Nature.* 2008;455:1061–1068.
- Brennan CW, Verhaak RGW, McKenna A; TCGA Research Network, et al. The somatic genomic landscape of glioblastoma. *Cell.* 2013;155:462–477.
- Parsons DW, Jones S, Zhang X, et al. An integrated genomic analysis of human glioblastoma multiforme. *Science.* 2008;321:1807–1812.
- Cancer Genome Atlas Research, Brat DJ, Verhaak RGW, Aldape KD, et al. Comprehensive, integrative genomic analysis of diffuse lower-grade gliomas. *N Engl J Med.* 2015;372:2481–2498.
- Reifenberger J, Reifenberger G, Liu L, et al. Molecular genetic analysis of oligodendroglial tumors shows preferential allelic deletions on 19q and 1p. *Am J Pathol.* 1994;145:1175–1190.
- Wu G, Broniscer A, McEachron TA, et al. Somatic histone H3 alterations in pediatric diffuse intrinsic pontine gliomas and non-brainstem glioblastomas. *Nat Genet.* 2012;44:251–253.
- Gessi M, Gielen GH, Hammes J, et al. H3.3 G34R mutations in pediatric primitive neuroectodermal tumors of central nervous system (CNS-PNET) and pediatric glioblastomas: possible diagnostic and therapeutic implications? *J Neurooncol.* 2013;112:67–72.
- Zhang J, Wu G, Miller CP, et al. Whole-genome sequencing identifies genetic alterations in pediatric low-grade gliomas. *Nat Genet.* 2013;45:602–612.
- Mackay A, Burford A, Carvalho D, et al. Integrated molecular meta-analysis of 1,000 pediatric high-grade and diffuse intrinsic pontine glioma. *Cancer Cell.* 2017;32:520–537 e5.
- Korshunov A, Schrimpf D, Ryzhova M, et al. H3-/IDH-wild type pediatric glioblastoma is comprised of molecularly and prognostically distinct subtypes with associated oncogenic drivers. *Acta Neuropathol.* 2017;134:507–516.
- Solomon DA, Korshunov A, Sill M, et al. Myxoid glioneuronal tumor of the septum pellucidum and lateral ventricle is defined by a recurrent PDGFRA p.K385 mutation and DNT-like methylation profile. *Acta Neuropathol.* 2018;136:339–343.
- Pekmezci M, Stevers M, Phillips JJ, et al. Multinodular and vacuolating neuronal tumor of the cerebrum is a clonal neoplasm defined by genetic alterations that activate the MAP kinase signaling pathway. *Acta Neuropathol.* 2018;135:485–488.
- Sturm D, Orr BA, Toprak UH, et al. New brain tumor entities emerge from molecular classification of CNS-PNETs. *Cell.* 2016;164:1060–1072.
- Pajtler KW, Witt H, Sill M, et al. Molecular classification of ependymal tumors across all CNS compartments, histopathological grades, and age groups. *Cancer Cell.* 2015;27:728–743.
- Northcott PA, Korshunov A, Witt H, et al. Medulloblastoma comprises four distinct molecular variants. *J Clin Oncol.* 2011;29:1408–1414.
- Sharma T, Schwalbe EC, Williamson D, et al. Second-generation molecular subgrouping of medulloblastoma: an international meta-analysis of group 3 and group 4 subtypes. *Acta Neuropathol.* 2019;138:309–326.
- Ho B, Johann PD, Grabovska Y, et al. Molecular subgrouping of atypical teratoid/rhabdoid tumors-a reinvestigation and current consensus. *Neuro Oncol.* 2020;22:613–624.
- Dean BL, Drayer BP, Bird CR, et al. Gliomas: classification with MR imaging. *Radiology.* 1990;174:411–415.
- Schwaighofer BW, Klein MV, Wesbey G, et al. Clinical experience with routine Gd-DTPA administration for MR imaging of the brain. *J Comput Assist Tomogr.* 1990;14:11–17.
- Aronen HJ, Gazit IE, Louis DN, et al. Cerebral blood volume maps of gliomas: comparison with tumor grade and histologic findings. *Radiology.* 1994;191:41–51.
- Tsuruda JS, Chew WM, Moseley ME, et al. Diffusion-weighted MR imaging of the brain: value of differentiating between extraaxial cysts and epidermoid tumors. *AJNR Am J Neuroradiol.* 1990;11:925–931. discussion 932–4.
- Maier SE, Sun Y, Mulkern RV. Diffusion imaging of brain tumors. *NMR Biomed.* 2010;23:849–864.
- Mueller WM, Yetkin FZ, Hammeke TA, et al. Functional magnetic resonance imaging mapping of the motor cortex in patients with cerebral tumors. *Neurosurgery.* 1996;39:515–521.
- Miloushev VZ, Granlund KL, Boltyskiy R, et al. Metabolic imaging of the human brain with hyperpolarized (¹³C) pyruvate demonstrates (¹³C) lactate production in brain tumor patients. *Cancer Res.* 2018;78:3755–3760.
- Wang ZJ, Ohliger MA, Larson PEZ, et al. Hyperpolarized (¹³C) MRI: state of the art and future directions. *Radiology.* 2019;291:273–284.
- Galldiks N, Lohmann P, Aboian M, et al. Update to the RANO working group and EANO recommendations for the clinical use of PET imaging in gliomas. *Lancet Oncol.* 2025;26:e436–e447.
- Shooli H, Nemati R, Ahmadzadehfah H, et al. Theranostics in brain tumors. *PET Clin.* 2021;16:397–418.
- Ellingson BM, Bendszus M, Boxerman J, et al. Consensus recommendations for a standardized brain tumor imaging protocol in clinical trials. *Neuro Oncol.* 2015;17:1188–1198.
- Sanvito F, Kaufmann TJ, Cloughesy TF, et al. Standardized brain tumor imaging protocols for clinical trials: current recommendations and tips for integration. *Front Radiol.* 2023;3:1267615.
- Kaufmann TJ, Smits M, Boxerman J, et al. Consensus recommendations for a standardized brain tumor imaging protocol for clinical trials in brain metastases. *Neuro Oncol.* 2020;22:757–772.
- Barajas RF, Politi LS, Anzalone N, et al. Consensus recommendations for MRI and PET imaging of primary Central nervous system lymphoma: guideline statement from the international primary CNS lymphoma collaborative group (IPCG). *Neuro Oncol.* 2021;23:1056–1071.
- Boxerman JL, Quarles CC, Hu LS, et al. Consensus recommendations for a dynamic susceptibility contrast MRI protocol for use in high-grade gliomas. *Neuro Oncol.* 2020;22:1262–1275.
- Wen PY, Macdonald DR, Reardon DA, et al. Updated response assessment criteria for high-grade gliomas: Response Assessment in Neuro-Oncology working group. *J Clin Oncol.* 2010;28:1963–1972.
- Wen PY, van den Bent M, Vogelbaum MA, et al. RANO 2.0: update to the response assessment in Neuro-Oncology criteria for high- and low-grade gliomas in adults. *Neuro Oncol.* 2024;26:2–4.
- Ellingson BM, Sanvito F, Cloughesy TF, et al. A neuroradiologist's guide to operationalizing the Response Assessment in Neuro-Oncology (RANO) criteria version 2.0 for gliomas in adults. *AJNR Am J Neuroradiol.* 2024;45:1846–1856.

40. Das S, Nayak GK, Saba L, et al. An artificial intelligence framework and its bias for brain tumor segmentation: a narrative review. *Comput Biol Med.* 2022;143:105273.
41. Ranjbarzadeh R, Caputo A, Tirkolaee EB, et al. Brain tumor segmentation of MRI images: a comprehensive review on the application of artificial intelligence tools. *Comput Biol Med.* 2023;152:106405.
42. Khalighi S, Reddy K, Midya A, et al. Artificial intelligence in neuro-oncology: advances and challenges in brain tumor diagnosis, prognosis, and precision treatment. *NPJ Precis Oncol.* 2024;8:80.
43. Stupp R, Mason WP, van den Bent MJ, et al. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. *N Engl J Med.* 2005;352:987–996.
44. van den Bent MJ, Tesileanu CMS, Wick W, et al. Adjuvant and concurrent temozolomide for 1p/19q non-co-deleted anaplastic glioma (CATNON; EORTC study 26053-22054): second interim analysis of a randomised, open-label, phase 3 study. *Lancet Oncol.* 2021;22:813–823.
45. Cairncross G, Wang M, Shaw E, et al. Phase III trial of chemoradiotherapy for anaplastic oligodendroglioma: long-term results of RTOG 9402. *J Clin Oncol.* 2013;31:337–343.
46. van den Bent MJ, Brandes AA, Taphoorn MJB, et al. Adjuvant procarbazine, lomustine, and vincristine chemotherapy in newly diagnosed anaplastic oligodendroglioma: long-term follow-up of EORTC brain tumor group study 26951. *J Clin Oncol.* 2013;31:344–350.
47. Friedman HS, Prados MD, Wen PY, et al. Bevacizumab alone and in combination with irinotecan in recurrent glioblastoma. *J Clin Oncol.* 2009;27:4733–4740.
48. Kreisl TN, Kim L, Moore K, et al. Phase II trial of single-agent bevacizumab followed by bevacizumab plus irinotecan at tumor progression in recurrent glioblastoma. *J Clin Oncol.* 2009;27:740–745.
49. Mellinghoff IK, van den Bent MJ, Blumenthal DT, et al. Vorasidenib in IDH1- or IDH2-mutant low-grade glioma. *N Engl J Med.* 2023;389:589–601.
50. Bouffet E, Hansford JR, Garrè ML, et al. Dabrafenib plus trametinib in pediatric glioma with BRAF V600 mutations. *N Engl J Med.* 2023;389:1108–1120.
51. Wen PY, Stein A, van den Bent M, et al. Dabrafenib plus trametinib in patients with BRAF(V600E)-mutant low-grade and high-grade glioma (ROAR): a multicentre, open-label, single-arm, phase 2, basket trial. *Lancet Oncol.* 2022;23:53–64.
52. Brastianos PK, Twohy E, Geyer S, et al. BRAF-MEK inhibition in newly diagnosed papillary craniopharyngiomas. *N Engl J Med.* 2023;389:118–126.
53. Kilburn LB, Khuong-Quang D-A, Hansford JR, et al. The type II RAF inhibitor tovorafenib in relapsed/refractory pediatric low-grade glioma: the phase 2 FIREFLY-1 trial. *Nat Med.* 2024;30:207–217.
54. Arrillaga-Romany I, Gardner SL, Odia Y, et al. ONC201 (dordaviprone) in recurrent H3 K27M-Mutant diffuse midline glioma. *J Clin Oncol.* 2024;42:1542–1552.
55. Gross AM, Wolters PL, Dombi E, et al. Selumetinib in children with inoperable plexiform neurofibromas. *N Engl J Med.* 2020;382:1430–1442.
56. Moertel CL, Hirbe AC, Shuhaiber HH, et al. ReNeu: a pivotal, phase IIb trial of mirdametininib in adults and children with symptomatic neurofibromatosis type 1-associated plexiform neurofibroma. *J Clin Oncol.* 2025;43:716–729.
57. Iliopoulos O, Iversen AB, Narayan V, et al. Belzutifan for patients with Von Hippel-Lindau disease-associated CNS haemangioblastomas (LITESPARK-004): a multicentre, single-arm, phase 2 study. *Lancet Oncol.* 2024;25:1325–1336.
58. Brastianos PK, Horowitz PM, Santagata S, et al. Genomic sequencing of meningiomas identifies oncogenic SMO and AKT1 mutations. *Nat Genet.* 2013;45:285–289.
59. Clark VE, Erson-Omay EZ, Serin A, et al. Genomic analysis of non-NF2 meningiomas reveals mutations in TRAF7, KLF4, AKT1, and SMO. *Science.* 2013;339:1077–1080.
60. Berghoff AS, Hielscher T, Ricken G, et al. Prognostic impact of genetic alterations and methylation classes in meningioma. *Brain Pathol.* 2022;32:e12970.
61. Chen WC, Choudhury A, Youngblood MW, et al. Targeted gene expression profiling predicts meningioma outcomes and radiotherapy responses. *Nat Med.* 2023;29:3067–3076.
62. Choudhury A, Chen WC, Lucas C-HG, et al. Hypermutic meningiomas harbor DNA methylation subgroups with distinct biological and clinical features. *Neuro Oncol.* 2023;25:520–530.
63. Landry AP, Wang JZ, Liu J, et al. Development and validation of a molecular classifier of meningiomas. *Neuro Oncol.* 2025;27:1258–1269.
64. Choudhury A, Magill ST, Eaton CD, et al. Meningioma DNA methylation groups identify biological drivers and therapeutic vulnerabilities. *Nat Genet.* 2022;54:649–659.
65. Sahm F, Schrimpf D, Stichel D, et al. DNA methylation-based classification and grading system for meningioma: a multicentre, retrospective analysis. *Lancet Oncol.* 2017;18:682–694.
66. Sahm F, Aldape KD, Brastianos PK, et al. cIMPACT-NOW update 8: clarifications on molecular risk parameters and recommendations for WHO grading of meningiomas. *Neuro Oncol.* 2025;27:319–330.
67. Brastianos PK, Galanis E, Butowski N, et al. Advances in multidisciplinary therapy for meningiomas. *Neuro Oncol.* 2019;21:i18–i31.
68. Du Z, Abedalthagafi M, Aizer AA, et al. Increased expression of the immune modulatory molecule PD-L1 (CD274) in anaplastic meningioma. *Oncotarget.* 2015;6:4704–4716.
69. Karimi S, Mansouri S, Mamatjan Y, et al. Programmed death ligand-1 (PD-L1) expression in meningioma: prognostic significance and its association with hypoxia and NFkB2 expression. *Sci Rep.* 2020;10:14115.
70. Mirian C, Duun-Henriksen AK, Maier A, et al. Somatostatin receptor-targeted radiopeptide therapy in treatment-refractory meningioma: individual patient data meta-analysis. *J Nucl Med.* 2021;62:507–513.
71. Vogelbaum MA, Brown PD, Messersmith H, et al. Treatment for brain metastases: ASCO-SNO-ASTRO guideline. *J Clin Oncol.* 2022;40:492–516.
72. Doebele RC, Drilon A, Paz-Ares L, et al. Entrectinib in patients with advanced or metastatic NTRK fusion-positive solid tumours: integrated analysis of three phase 1-2 trials. *Lancet Oncol.* 2020;21:271–282.
73. Reungwetwattana T, Nakagawa K, Cho BC, et al. CNS response to osimertinib versus standard epidermal growth factor receptor tyrosine kinase inhibitors in patients with untreated EGFR-Mutated advanced non-small-cell lung cancer. *J Clin Oncol.* 2018;36:Jco2018783118.
74. Sabari JK, Velcheti V, Shimizu K, et al. Activity of adagrasib (MRTX849) in brain metastases: preclinical models and clinical data from patients with KRASG12C-mutant non-small cell lung cancer. *Clin Cancer Res.* 2022;28:3318–3328.
75. Shaw AT, Bauer TM, de Marinis F, et al. First-Line lorlatinib or crizotinib in advanced ALK-positive lung cancer. *N Engl J Med.* 2020;383:2018–2029.
76. Murthy RK, Loi S, Okines A, et al. Tucatinib, trastuzumab, and capecitabine for HER2-positive metastatic breast cancer. *N Engl J Med.* 2020;382:597–609.
77. Davies MA, Saiag P, Robert C, et al. Dabrafenib plus trametinib in patients with BRAF(V600)-mutant melanoma brain metastases (COMBI-MB): a multicentre, multicohort, open-label, phase 2 trial. *Lancet Oncol.* 2017;18:863–873.
78. Long GV, Atkinson V, Lo S, et al. Combination nivolumab and ipilimumab or nivolumab alone in melanoma brain metastases: a multicentre randomised phase 2 study. *Lancet Oncol.* 2018;19:672–681.
79. Tawbi HA, Forsyth PA, Algazi A, et al. Combined nivolumab and ipilimumab in melanoma metastatic to the brain. *N Engl J Med.* 2018;379:722–730.

80. Brastianos PK, Carter SL, Santagata S, et al. Genomic characterization of brain metastases reveals branched evolution and potential therapeutic targets. *Cancer Discov*. 2015;5:1164–1177.
81. Hulsbergen AFC, Claes A, Kavouridis VK, et al. Subtype switching in breast cancer brain metastases: a multicenter analysis. *Neuro Oncol*. 2020;22:1173–1181.
82. Wen PY, Packer RJ. The 2021 WHO classification of tumors of the central nervous system: clinical implications. *Neuro Oncol*. 2021;23:1215–1217.
83. Pratt D, Sahm F, Aldape K. DNA methylation profiling as a model for discovery and precision diagnostics in neuro-oncology. *Neuro Oncol*. 2021;23:S16–s29.
84. Gupta N, Goumnerova LC, Manley P, et al. Prospective feasibility and safety assessment of surgical biopsy for patients with newly diagnosed diffuse intrinsic pontine glioma. *Neuro Oncol*. 2018;20:1547–1555.
85. Franson A, Koschmann C. Enhancing GD2 CAR T-cell therapy with IGF-1R blockade: are DIPG CAR T cells ready for combinatorial therapy? *Neuro Oncol*. 2022;24:1164–1165.
86. Miller AM, Szalontay L, Bouvier N, et al. Next-generation sequencing of cerebrospinal fluid for clinical molecular diagnostics in pediatric, adolescent and young adult brain tumor patients. *Neuro Oncol*. 2022;24:1763–1772.
87. Cooney TM, Cohen KJ, Guimaraes CV, et al. Response assessment in diffuse intrinsic pontine glioma: recommendations from the response assessment in Pediatric Neuro-Oncology (RAPNO) working group. *Lancet Oncol*. 2020;21:e330–e336.
88. Tsai JW, Choi JJ, Ouassalam H, et al. Integrated response analysis of pediatric low-grade gliomas during and after targeted therapy treatment. *Neurooncol Adv*. 2023;5:vdac182.
89. Warren KE, Vezina G, Poussaint TY, et al. Response assessment in medulloblastoma and leptomeningeal seeding tumors: recommendations from the response assessment in pediatric Neuro-Oncology committee. *Neuro Oncol*. 2018;20:13–23.
90. Fangusaro J, Jones DT, Packer RJ, et al. Pediatric low-grade glioma: state-of-the-art and ongoing challenges. *Neuro Oncol*. 2024;26:25–37.
91. de la Fuente MI, Touat M, van den Bent MJ, et al. The role of vorasidenib in the treatment of isocitrate dehydrogenase-mutant glioma. *Neuro Oncol*. 2024;27:1135–1148.
92. Rosenberg T, Yeo KK, Mauguén A, et al. Upfront molecular targeted therapy for the treatment of BRAF-mutant pediatric high-grade glioma. *Neuro Oncol*. 2022;24:1964–1975.
93. Ziegler DS, Lehmann R, Eisenstat DD. A paradigm shift in how we treat pediatric low-grade glioma-Targeting the molecular drivers. *Neuro Oncol*. 2024;26:593–595.
94. Tseng YD, Hartsell W, Tsai H, et al. Proton therapy patterns of care among pediatric and adult patients with CNS tumors. *Neuro Oncol*. 2018;20:1556–1557.
95. Leeper HE, Acquaye AA, Bell S, et al. Survivorship care planning in neuro-oncology. *Neurooncol Pract*. 2018;5:3–9.
96. Lim M, Weller M, Idbaih A, et al. Phase III trial of chemoradiotherapy with temozolomide plus nivolumab or placebo for newly diagnosed glioblastoma with methylated MGMT promoter. *Neuro Oncol*. 2022;24:1935–1949.
97. Omuro A, Brandes AA, Carpentier AF, et al. Radiotherapy combined with nivolumab or temozolomide for newly diagnosed glioblastoma with unmethylated MGMT promoter: an international randomized phase III trial. *Neuro Oncol*. 2023;25:123–134.
98. Reardon DA, Brandes AA, Omuro A, et al. Effect of nivolumab vs bevacizumab in patients with recurrent glioblastoma: the CheckMate 143 phase 3 randomized clinical trial. *JAMA Oncol*. 2020;6:1003–1010.
99. Nassiri F, Patil V, Yefet LS, et al. Oncolytic DNX-2401 virotherapy plus pembrolizumab in recurrent glioblastoma: a phase 1/2 trial. *Nat Med*. 2023;29:1370–1378.
100. Desjardins A, Gromeier M, Herndon JE, et al. Recurrent glioblastoma treated with recombinant poliovirus. *N Engl J Med*. 2018;379:150–161.
101. Ling AL, Solomon IH, Landivar AM, et al. Clinical trial links oncolytic immunoactivation to survival in glioblastoma. *Nature*. 2023;623:157–166.
102. Galanis E, Dooley KE, Keith Anderson S, et al. Carcinoembryonic antigen-expressing oncolytic measles virus derivative in recurrent glioblastoma: a phase 1 trial. *Nat Commun*. 2024;15:493.
103. Mendez-Gomez HR, DeVries A, Castillo P, et al. RNA aggregates harness the danger response for potent cancer immunotherapy. *Cell*. 2024;187:2521–2535.e21.
104. Reardon DA, Brem S, Desai AS, et al. Intramuscular (IM) INO-5401 + INO-9012 with electroporation (EP) in combination with cemiplimab (REGN2810) in newly diagnosed glioblastoma. *JCO*. 2022;40:2004–2004.
105. Brown CE, Hibbard JC, Alizadeh D, et al. Locoregional delivery of IL-13R α 2-targeting CAR-T cells in recurrent high-grade glioma: a phase 1 trial. *Nat Med*. 2024;30:1001–1012.
106. Choi BD, Gerstner ER, Frigault MJ, et al. Intraventricular CARv3-TEAM-E T cells in recurrent glioblastoma. *N Engl J Med*. 2024;390:1290–1298.
107. Bagley SJ, Logun M, Fraietta JA, et al. Intrathecal bivalent CAR T cells targeting EGFR and IL13R α 2 in recurrent glioblastoma: phase 1 trial interim results. *Nat Med*. 2024;30:1320–1329.
108. Gallus M, Yamamichi A, Arrieta VA, et al. Abstract 3527: low-intensity pulsed focused ultrasound, with microbubbles and immune adjuvants, promotes CNS epitope spreading, improving CAR T cell therapy for heterogeneous brain tumors. *Cancer Res*. 2025;85:3527–3527.
109. Sabbagh A, Beccaria K, Ling X, et al. Opening of the blood-brain barrier using low-intensity pulsed ultrasound enhances responses to immunotherapy in preclinical glioma models. *Clin Cancer Res*. 2021;27:4325–4337.
110. Shumer-Elbaz M, Ad-El N, Chulanova Y, et al. Low-frequency ultrasound-mediated blood-brain barrier opening enables non-invasive lipid nanoparticle RNA delivery to glioblastoma. *J Control Release*. 2025;385:114018.
111. Wang M-F, Guo J, Yuan S-J, et al. Targeted sonodynamic therapy induces tumor cell quasi-immunogenic ferroptosis and macrophage immunostimulatory autophagy in glioblastoma. *Biomaterials*. 2025;315:122913.
112. Zhou Y, Jiao J, Yang R, et al. Temozolomide-based sonodynamic therapy induces immunogenic cell death in glioma. *Clin Immunol*. 2023;256:109772.
113. Arrieta VA, Gould A, Kim K-S, et al. Ultrasound-mediated delivery of doxorubicin to the brain results in immune modulation and improved responses to PD-1 blockade in gliomas. *Nat Commun*. 2024;15:4698.
114. Buonfiglioli A, Hambardzumyan D. Macrophages and microglia: the cerberus of glioblastoma. *Acta Neuropathol Commun*. 2021;9:54.
115. Cordell EC, Alghamri MS, Castro MG, et al. T lymphocytes as dynamic regulators of glioma pathobiology. *Neuro Oncol*. 2022;24:1647–1657.
116. Song KW, Lim M, Monje M. Complex neural-immune interactions shape glioma immunotherapy. *Immunity*. 2025;58:1140–1160.
117. Anastasaki C, Gao Y, Gutmann DH. Neurons as stromal drivers of nervous system cancer formation and progression. *Dev Cell*. 2023;58:81–93.
118. Winkler F, Venkatesh HS, Amit M, et al. Cancer neuroscience: state of the field, emerging directions. *Cell*. 2023;186:1689–1707.
119. Ross JL, Puigdelloses-Vallcorba M, Piñero G, et al. Microglia and monocyte-derived macrophages drive progression of pediatric high-grade gliomas and are transcriptionally shaped by histone mutations. *Immunity*. 2024;57:2669–2687.e6.
120. Tripathi S, Najem H, Dussold C, et al. Pediatric glioma immune profiling identifies TIM3 as a therapeutic target in BRAF fusion pilocytic astrocytoma. *J Clin Invest*. 2024;134.
121. Zhang L, et al. A zinc transporter drives glioblastoma progression via extracellular vesicles-reprogrammed microglial plasticity. *Proc Natl Acad Sci U S A*. 2025;122:e2427073122.

122. Akkari L, Bowman RL, Tessier J, et al. Dynamic changes in glioma macrophage populations after radiotherapy reveal CSF-1R inhibition as a strategy to overcome resistance. *Sci Transl Med*. 2020;12:
123. Chen Z, et al. A paracrine circuit of IL-1 β /IL-1R1 between myeloid and tumor cells drives genotype-dependent glioblastoma progression. *J Clin Invest*. 2023;133:
124. Pyonteck SM, Akkari L, Schuhmacher AJ, et al. CSF-1R inhibition alters macrophage polarization and blocks glioma progression. *Nat Med*. 2013;19:1264–1272.
125. Watson SS, Zomer A, Fournier N, et al. Fibrotic response to anti-CSF-1R therapy potentiates glioblastoma recurrence. *Cancer Cell*. 2024;42:1507–1527.e11.
126. Ye Z, Ai X, Yang K, et al. Targeting microglial metabolic rewiring synergizes with Immune-Checkpoint blockade therapy for glioblastoma. *Cancer Discov*. 2023;13:974–1001.
127. Barakat R, Chatterjee J, Mu R, et al. Human single cell RNA-sequencing reveals a targetable CD8(+) exhausted T cell population that maintains mouse low-grade glioma growth. *Nat Commun*. 2024;15:10312.
128. Friebe E, Kapolou K, Unger S, et al. Single-Cell mapping of human brain cancer reveals Tumor-Specific instruction of Tissue-Invasive leukocytes. *Cell*. 2020;181:1626–1642.e20.
129. Das A, Sudhaman S, Morgenstern D, et al. Genomic predictors of response to PD-1 inhibition in children with germline DNA replication repair deficiency. *Nat Med*. 2022;28:125–135.
130. de Andrade Costa A, Chatterjee J, Cobb O, et al. Immune deconvolution and temporal mapping identifies stromal targets and developmental intervals for abrogating murine low-grade optic glioma formation. *Neurooncol Adv*. 2022;4:vdab194.
131. Park J, Kwon M, Kim KH, et al. Immune checkpoint inhibitor-induced reinvigoration of tumor-infiltrating CD8(+) T cells is determined by their differentiation status in glioblastoma. *Clin Cancer Res*. 2019;25:2549–2559.
132. Anastasaki C, Chatterjee J, Koleske JP, et al. NF1 mutation-driven neuronal hyperexcitability sets a threshold for tumorigenesis and therapeutic targeting of murine optic glioma. *Neuro Oncol*. 2024;26:1496–1508.
133. Anastasaki C, Mo J, Chen J-K, et al. Neuronal hyperexcitability drives central and peripheral nervous system tumor progression in models of neurofibromatosis-1. *Nat Commun*. 2022;13:2785.
134. Pan Y, Hysinger JD, Barron T, et al. NF1 mutation drives neuronal activity-dependent initiation of optic glioma. *Nature*. 2021;594:277–282.
135. Venkatesh HS, Tam LT, Woo PJ, et al. Targeting neuronal activity-regulated neuroligin-3 dependency in high-grade glioma. *Nature*. 2017;549:533–537.
136. Venkataramani V, Yang Y, Ille S, et al. Cancer neuroscience of brain tumors: from multicellular networks to Neuroscience-Instructed cancer therapies. *Cancer Discov*. 2025;15:39–51.